

Matplotlib Exercises

1. Orbiting Particle

Write a program with a Particle class that plots the position of the particle as it orbits a central point under the influence of an inverse square force (F).

Define a the class along the lines:

```
class Particle:
    def __init__(self, name, x0, v0): ...
    def getPosition(self): ...
    def next(self, dt): ...
```

where x_0 and v_0 are the initial position and velocity of the particle. The *next* method calculates the new position (x) and velocity (v) of the particle a time interval dt later.

Call the next method repeatedly and use the formulae:

```
dv = F * dt / m
dx = v * dt
```

to calculate the new position (x) and velocity (v) of the particle:

```
v = v + dv
x = x + dx
```

Plot the resulting orbit in 3D with matplotlib.

